

CHRISTOPHER AVINA

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OBJECTIVE

Results driven and adaptable engineer with hands-on experience in software development and data engineering. I am proficient in Python, C++, Java, and JavaScript, with a proven track record of solving complex problems and adapting to new technologies. I am eager to apply my technical skillset and passion for data-driven solutions to contribute to innovative projects.

EXPERIENCE

DATA ENGINEER, Freelance (Python/Databricks/PostgreSQL/Pyspark/AWS) [Github Repo](#) *August 2024-Present*

- Extracted and stored data from NASA API in a Datalake with medallion architecture.
- Used AWS S3 buckets to store the data in appropriate formats with respect to the pipeline stage.
- Loaded, transformed and normalized data tables with Pyspark.
- Loaded normalized data from silver and created interactive dashboards using Databricks, Numpy, Pyplotlib, and OpenCV.
- Automated the ETL process with Workflows, reducing manual intervention and improving data handling efficiency.

EMBEDDED SOFTWARE ENGINEER, Rekovar (C/C++/Java) *April 2023 - June 2023*

Left due to cancer treatments. Now cancer free!

- Developed and debugged neonatal medical device with offshore engineering team.
- Led a comprehensive overhaul of project structure and build systems, fostering a test-centric codebase and increasing engineer productivity by 10%.
- Collaborated with offshore team to track, debug, and solve protocols such as I2C and BLE data loss.
- Developed comprehensive unit tests using Ceedling and Unity, utilizing CMock to mock drivers, increasing test code coverage from 0 to 100%.
- Created documentation and SOPs for the QMS system, Greenlight Guru, which actively contributed to the adherence of FDA approval guidelines.

PROJECTS

Starcraft 2 Reinforcement Learning AI (Python/Pandas/Numpy) [Github Repo](#) *Jan 2025-Present*

- Developed a Zerg AI in Python, leveraging reinforcement learning with a Q-learning table for strategic decision-making.
- Utilized Deepmind's PySC2 to extract features from the game environment allowing for analysis of the decision-making process.
- Tuned reward functions to reinforce optimal behaviors, including efficient drone production, tech progression, and combat engagements.
- Optimized AI performance through experience-driven updates and training iterations, improving macro and micro decision-making.

Encrypted Baby Monitor (Python/Jetson Nano/Flask/SKlearn/OpenCV) [Github Repo](#) *Sep 2022- March 2023*

- Led a team of engineers at UCI to develop an edge-computing media server using a Jetson Nano, which captures live video footage and analyzes it using various machine learning models to determine if a child is in a dangerous position.
- Trained multiple models using SKLearn on a custom data set built using dolls.
- Deployed best rated model to the media-server, successfully detecting unwanted positions with a 70% success rate.
- Implemented AES-128 encryption on the device to protect the video footage from unauthorized access or tampering.
- Created a Flask application that connected to the media-server and displayed the video stream and data in real time.

EDUCATION

UNIVERSITY OF CALIFORNIA, IRVINE

B.S. Computer Science & Engineering

Spring 2023

SKILLS

LANGUAGES

Python, C/C#/C++, Java, Javascript, SQL

FRAMEWORKS/LIBRARIES

SKLearn, Tensorflow, OpenCV, Flask, Django, MERN Stack, Numpy, Pyplotlib, SDL2, Unreal Engine, Pygame

DATA & CLOUD

Databricks, AWS, Snowflake, Spark, Airflow, PostgreSQL, MySQL, MongoDB, PySpark

EMBEDDED

ARM/RISC-V Architecture (STM32, ATmega32x, ESP32), Systems Communication (I2C, SPI, UART, BLE), PCB Design (Kicad), Verilog (HDL), Circuit Design & Analysis, Root Cause Analysis